

JEE ADVANCED - 4 | PAPER - 1

JEE 2022

Date	Maximum Marks	Duration
10th October, 2021	198	3 Hours

General Instructions

- The question paper consists of 3 Subject (Subject I: **Physics**, Subject II: **Chemistry**, Subject III: **Mathematics**). Each Part has **three** sections (Section 1, Section 2 & Section 3).
- Section 1** contains **6 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

Section 2 contains **6 Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

Section 3 contains **6 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
- For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code**, **Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

MARKING SCHEME

SECTION - 1 (Maximum Marks: 18)

- This section contains **SIX (06)** Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If only (all) the correct option(s) is(are) chosen
Zero Mark: 0 if none of the options is chosen (i.e. the question is unanswered)
Negative Marks: -1 In all other cases.

SECTION - 2 (Maximum Marks: 24)

- This section consists of **SIX (06)** Questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If only (all) the correct option(s) is(are) chosen
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct
Partial Marks: +1 If two or more options are correct but **ONLY** one option is chosen, and it is a correct option
Zero Mark: 0 if none of the options is chosen (i.e. the question is unanswered)
Negative Marks: -2 In all other cases.

SECTION – 3 (Maximum Marks: 24)

- This section contains **6 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If **ONLY** the correct Integer value is entered. There is **NO negative marking**.
Zero Marks: 0 In all other cases.

SECTION 1

SINGLE CORRECT ANSWERS TYPE

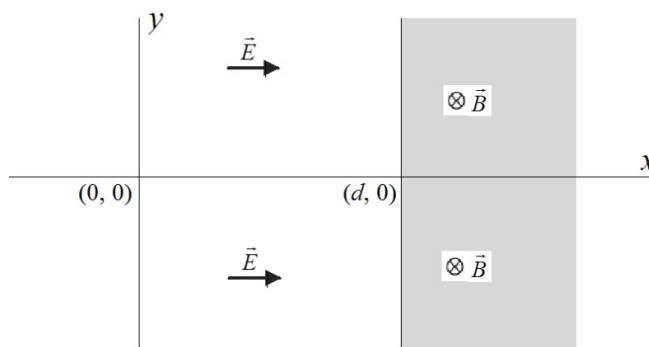
This section contains 06 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- A rectangular wire loop of side lengths a and b carries a steady current I . The magnetic field intensity at the centre of the rectangle (i.e. the point of intersection of its diagonals) is:

(A) $\frac{2\mu_0 I}{\pi} \left(\frac{\sqrt{a^2 + b^2}}{ab} \right)$ (B) $\frac{4\mu_0 I}{\pi} \left(\frac{\sqrt{a^2 + b^2}}{ab} \right)$

(C) $\frac{4\sqrt{2}\mu_0 I}{\pi} \left(\frac{\sqrt{ab}}{a^2 + b^2} \right)$ (D) $\frac{8\sqrt{2}\mu_0 I}{\pi} \left(\frac{\sqrt{ab}}{a^2 + b^2} \right)$
- A cylindrical shell of inner and outer radius R and $3R$ and length L is made out of a material of resistivity ρ . A potential difference V is applied between the inner and outer curved surfaces of the shell. The electric field at a point at a distance $2R$ from the axis of the shell is:

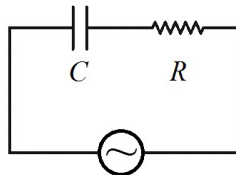
(A) $\frac{V}{R \log_e(3)}$ (B) $\frac{V}{2R \log_e(3)}$ (C) $\frac{V}{R \log_e(2)}$ (D) $\frac{V}{2R \log_e(2)}$
- A uniform electric field of magnitude E pointing in the $+X$ direction exists in the region $0 \leq x < d$, and a uniform magnetic field of magnitude B pointing in the $-Z$ direction exists in the region $x \geq d$. A point particle of mass m and carrying positive charge q is released from rest at the origin. The Y -coordinate of the point where the particle leaves the magnetic field for the first time is y_0 . Then, y_0 is proportional to:



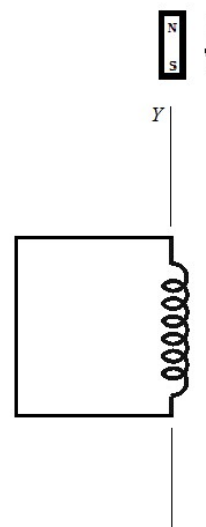
- (A) $\left(\frac{Ed^2}{mqB^2} \right)^{1/2}$ (B) $\left(\frac{mEd}{qB^2} \right)^{1/2}$ (C) $\left(\frac{mqEd}{B^2} \right)^{1/2}$ (D) $\left(\frac{mE}{qB^2d} \right)^{1/2}$
- A square wire loop of side length L is made out of a uniform wire of resistance per unit length β . The wire loop is made to rotate with a constant angular velocity ω about an axis in the plane of the loop, passing through its centre and parallel to two of its sides. If a uniform magnetic field of magnitude B along a direction perpendicular to the axis of rotation of the loop is switched on, the maximum value of the induced current in the loop is: (Neglect the self-inductance of the loop)

(A) $\frac{BL\omega}{2\beta}$ (B) $\frac{BL\omega}{4\beta}$ (C) $\frac{B\omega}{2\beta L}$ (D) $\frac{B\omega}{4\beta L}$

5. A capacitor of capacitance C is connected in series with a resistance R and a source of alternating voltage of angular frequency $\omega = \frac{1}{RC}$. The current in the circuit varies with time as: $I(t) = I_0 \cos(\omega t)$, with the clockwise direction of current taken as positive. At $t=0$, the capacitor is completely uncharged. The charge on the left plate of the capacitor varies with time as:



- (A) $Q(t) = \frac{I_0}{\omega} \sin(\omega t)$ (B) $Q(t) = \frac{I_0}{\omega} \cos\left(\omega t - \frac{\pi}{4}\right)$
 (C) $Q(t) = \frac{\sqrt{2}I_0}{\omega} \sin(\omega t)$ (D) $Q(t) = \frac{\sqrt{2}I_0}{\omega} \cos\left(\omega t - \frac{\pi}{4}\right)$
6. A solenoid is fixed with its axis along the Y-axis as shown. A bar magnet placed along the Y-axis and initially at a very large distance from the solenoid is moved at a constant velocity in the -Y direction. Until the magnet is again at a very large distance from the solenoid, the current induced in the solenoid:
- (A) is constant in magnitude but reverses direction once
 (B) is in the same direction throughout, but becomes zero once instantaneously
 (C) reverses direction once, becomes zero once instantaneously and reaches its maximum magnitude twice
 (D) is in the same direction throughout, and reaches its maximum magnitude once



SPACE FOR ROUGH WORK

SECTION 2

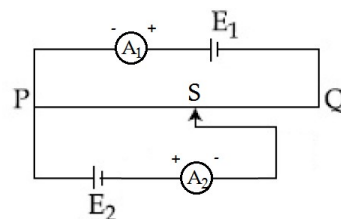
MULTIPLE CORRECT ANSWERS TYPE

This section consists of 06 Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

7. A circular loop of radius a is made out of a uniform wire of diameter d (such that $d \ll a$) and resistivity ρ . The loop is fixed in the X-Y plane, in the presence of a magnetic field $\vec{B} = B_0 \sin(\omega t) \hat{k}$, where B_0 and ω are constants. Let the maximum magnitude of the induced current in the loop be I_M and let the average rate of heat dissipation in the loop be H_{AV} . Which of these options is/are correct?
- (A) $I_M = \frac{\pi B_0 \omega a d^2}{8\rho}$ (B) $I_M = \frac{\pi B_0 \omega a^2 d}{8\rho}$
- (C) $H_{AV} = \frac{\pi^2 B_0^2 \omega^2 a d^3}{16\rho}$ (D) $H_{AV} = \frac{\pi^2 B_0^2 \omega^2 a^3 d^2}{16\rho}$
8. Two long insulating wires are placed parallel to each other in the X-Y plane, along the lines $x = -a$ and $x = a$ respectively. The wires are uniformly charged with linear charge densities $-\lambda$ and λ respectively. Which of these options is/are correct?
- (A) At any point on the Z-axis, the electric field is in the $-X$ direction
- (B) The Y-Z plane is an equipotential surface
- (C) The electric field at the point $(x, 0, 0)$ [$0 < x < a$] is $\left[\frac{\lambda x}{\pi \epsilon_0 (a^2 + x^2)} \right] (-\hat{i})$
- (D) The work done in taking a point charge $+q$ from $(0, 0, 0)$ to $\left(\frac{a}{2}, 0, 0\right)$ is $\frac{\lambda q}{2\pi \epsilon_0} \log_e 3$
9. A series RLC circuit consisting of an unknown resistance R , a capacitor of capacitance $C = 10 \mu\text{F}$ and a coil of inductance $L = 1 \text{ H}$ is connected to an AC voltage source of angular frequency $\omega = 100 \text{ rad/s}$. It is found that the current in the circuit leads the voltage source by a phase angle of $\frac{\pi}{4}$ and the average power dissipated in the resistance is 90 W . Which of these options is/are correct?
- (A) The rms value of the current in the circuit is $\frac{1}{\sqrt{10}} \text{ A}$
- (B) The power factor is $\frac{1}{\sqrt{2}}$
- (C) The amplitude of the voltage source is 180 V
- (D) The current in the circuit lags behind the voltage source by a phase angle of $\frac{\pi}{4}$ if the angular frequency of the source is changed to 1000 rad/s

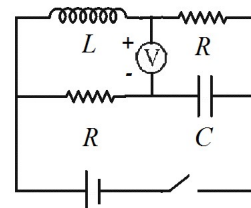
10. In the given circuit, PQ is a uniform potentiometer wire of length 120 cm and resistance $40\ \Omega$. The length PS is 90 cm. The EMF of the battery E_2 is 20 V. The battery E_1 is ideal, but the battery E_2 is non-ideal. The ideal ammeters A_1 and A_2 read +1.8 A and +1.0 A respectively. Which of these options is/are correct?

- (A) The potential difference between P and S is 24 V
 (B) The EMF of the battery E_1 is 36 V
 (C) The internal resistance of the battery E_2 is $4\ \Omega$
 (D) The current through ammeter A_2 can be made zero by changing the length PS

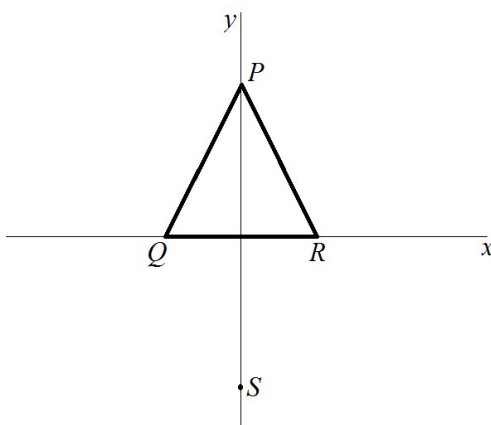


11. An uncharged capacitor of capacitance $C = 10\ \mu\text{F}$ and an inductor coil of inductance $L = 2\ \text{mH}$ are connected with two identical resistances $R = 20\ \Omega$ each, an ideal voltmeter and an ideal battery of emf 100 V as shown. Initially, the switch is open and current is zero everywhere. The switch is closed at $t = 0$. If the reading of the voltmeter reaches its maximum value V_{MAX} at $t = \tau$, then:

- (A) $\tau = (2\log_e(2)) \times 10^{-4}\ \text{s}$ (B) $V_{MAX} = 25\ \text{V}$
 (C) $\tau = (4\log_e(2)) \times 10^{-4}\ \text{s}$ (D) $V_{MAX} = 50\ \text{V}$



12. A triangular, equilateral loop is placed in the X-Y plane with its vertices at $P\left(0, \frac{a\sqrt{3}}{2}\right)$, $Q\left(-\frac{a}{2}, 0\right)$ and $R\left(\frac{a}{2}, 0\right)$. The loop carries a steady current I in the direction $P \rightarrow Q \rightarrow R \rightarrow P$. Let the point S have the coordinates $\left(0, -\frac{a\sqrt{3}}{2}\right)$. Which of the following statements is/are correct?

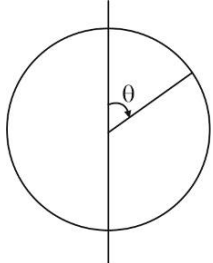


- (A) The magnetic field at S due to the side PQ is in the $-Z$ direction
 (B) The magnetic field at S due to the side PQ and the magnetic field at S due to the side RP have the same magnitude
 (C) The net magnetic field at S is in the $-Z$ direction
 (D) The net magnetic field at S has magnitude $\frac{\mu_0 I}{2\pi a} \left(\frac{2}{\sqrt{3}} - 1 \right)$

SECTION 3

NUMERICAL VALUE TYPE QUESTIONS

This section has 06 Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25)

13. A square wire loop of side length L is made out of a uniform thin conducting wire. The resistance of the loop is R . The loop is placed in the X-Y plane with its sides parallel to the coordinate axes and its centre at the origin. The loop is pivoted to rotate without friction about the Y-axis. A uniform magnetic field of magnitude B exists in the +Z direction. At $t = 0$, the loop is given an angular impulse J about the Y-axis. If the loop rotates by an angle 180° before it comes to rest, and $J = X \left(\frac{B^2 L^4}{R} \right)$, then X is _____.
14. Electric field at a point (x, y, z) in space is given by $\vec{E}(x, y, z) = x^2 \hat{i} - y^2 \hat{j} + 2z \hat{k}$ in SI units.. Charge density at a point $(2, 1, 1)$ is $P \epsilon_0$, where ϵ_0 is permittivity of free space. Value of P is _____.
15. A metal rod of length $L = 10$ cm pivoted at one end rotates in the horizontal plane at a constant rate of $\frac{240}{\pi}$ rotations per minute. If a uniform magnetic field of intensity $B = 2$ T is switched on in the vertical direction, the potential difference (in Volts) that develops across the ends of the rod is _____.
16. An uncharged metallic sphere of radius R is placed into an external uniform field, as a result of which an induced charge appears on the sphere with surface density $\sigma = \sigma_0 \cos \theta$, σ_0 is a positive constant and θ is the polar angle. Magnitude of resultant electric force acting on the positive charges on the sphere is $\frac{\alpha \sigma_0^2 R^2 \pi}{\epsilon_0}$, value of α is _____. 
17. Two identical capacitors of plate area A and plate separation d initially have vacuum between the plates. Now, two dielectric slabs of dielectric constant K and $4K$ and widths $\frac{d}{3}$ and $\frac{2d}{3}$ are inserted in the first capacitor, and a single dielectric slab of dielectric constant K_0 and width d is inserted into the second capacitor. After the insertion, the capacitance of the two capacitors is equal. Then, $\frac{K_0}{K} = \underline{\hspace{2cm}}$.
18. A large planar, thin insulating sheet is placed parallel to the horizontal and given a uniform charge density $\sigma = 2 \times 10^{-4} \text{ C/m}^2$. A small dipole of dipole moment $|\vec{p}| = 5 \times 10^{-7} \text{ Cm}$ is fixed above the sheet such that the dipole moment vector points in the vertically upward direction. The work done (in Joules) in slowly rotating the dipole such that its dipole moment vector now points in the vertically downward direction is _____. $[\epsilon_0 = \frac{80}{9} \times 10^{-12} \text{ C}^2 \text{N}^{-2} \text{m}^{-2}]$

SPACE FOR ROUGH WORK

SUBJECT II : CHEMISTRY**66 MARKS****SECTION 1****SINGLE CORRECT ANSWERS TYPE**

This section contains 06 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Liquids A and B form an ideal solution. The plot of $\frac{1}{X_A}$ (Y-axis) versus $\frac{1}{Y_A}$ (X-axis) (where X_A and Y_A are the mole fractions of A in liquid and vapour phases at equilibrium respectively) is linear whose Y-intercept and slope, respectively, are given as:

(A) $\frac{P_A^\circ - P_B^\circ}{P_B^\circ}, \frac{P_A^\circ}{P_B^\circ}$ (B) $\frac{P_A^\circ}{P_B^\circ}, \frac{P_B^\circ - P_A^\circ}{P_B^\circ}$ (C) $\frac{P_B^\circ - P_A^\circ}{P_B^\circ}, \frac{P_A^\circ}{P_B^\circ}$ (D) $\frac{P_B^\circ}{P_A^\circ}, \frac{P_B^\circ - P_A^\circ}{P_B^\circ}$

2. When 2g non volatile hydrocarbon containing 94.4% carbon by mass, is dissolved in 100g benzene, the vapour pressure of benzene at 30°C is lowered from 89.78 mm to 89 mm. The molecular formula of hydrocarbon is:

(A) $C_{12}H_{34}$ (B) $C_{13}H_{22}$ (C) $C_{14}H_{12}$ (D) $C_{14}H_{10}$

3. $SO_2Cl_2 \longrightarrow SO_2 + Cl_2$ is a first order gaseous reaction, with rate constant $k = 2.5 \times 10^{-5} \text{ sec}^{-1}$ at 320°C. The percentage of SO_2Cl_2 decomposed on heating for 100 minutes is? [$\ln(1.16) = 0.15$]

(A) 86.2% (B) 15% (C) 85% (D) 13.8%

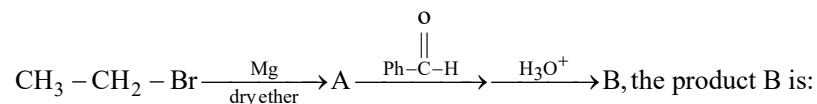
4. At 70K, the adsorption of $N_2(g)$ at iron surface obeys Freundlich adsorption isotherm. The following data is collected experimentally:

P(bar)	4	25	64
$\frac{x}{m}$	0.2	0.5	0.8

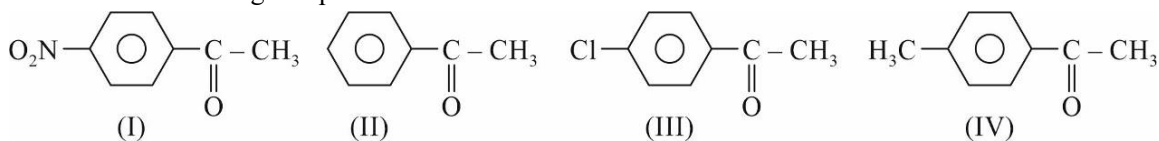
Where $\frac{x}{m}$ is the mass (in g) of $N_2(g)$ adsorbed per g of iron at 'P' bar pressure. The number of moles of $N_2(g)$ adsorbed per g iron at 36 bar and 70K is:

(A) $\frac{3}{10}$ (B) $\frac{3}{70}$ (C) $\frac{3}{140}$ (D) $\frac{3}{280}$

5. In the reaction



6. Consider the following compounds.



Choose the correct order of reactivity for nucleophilic addition reaction:

- (A) I > II > III > IV (B) IV > II > III > I
 (C) I > IV > III > II (D) I > III > II > IV

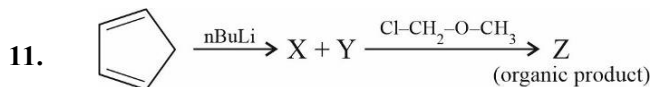
SPACE FOR ROUGH WORK

SECTION 2

MULTIPLE CORRECT ANSWERS TYPE

This section consists of 06 Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

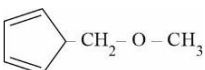
7. Select the correct statements about three dimensional HCP system:
- (A) Number of atoms in HCP unit cell is 6
 (B) The volume of HCP unit cell is $24\sqrt{3}r^3$
 (C) The base area of HCP unit cell is $6\sqrt{3}r^2$
 (D) The empty space in HCP unit cell is 26%
8. Which of the following statements is/are correct:
- (A) For endothermic reactions, activation energy is greater than heat of reaction
 (B) Rate of elementary reaction always increases with increase in temperature
 (C) For exothermic reactions, activation energy is always smaller than heat of reaction
 (D) Temperature coefficient (μ) of a reaction is 2 when $T \rightarrow \infty$
9. The incorrect statement(s) is/are:
- (A) Tyndall effect is more effective in gold sol in comparison to the rubber sol
 (B) The conductivity of a soap solution decreases sharply at CMC
 (C) CMC value of $\text{CH}_3(\text{CH}_2)_9\text{NH}_3^+\text{Cl}^-$ will be lesser than that of $\text{CH}_3(\text{CH}_2)_{12}\text{COONa}$
 (D) The elevation in boiling point of an alcoholic solution of sulphur is less than that of its sol in water if mass of sulphur present per unit volume of mixture and K_b is same in both cases
10. The reactions taking place in the dry cell are:
- Anode: $\text{Zn} \longrightarrow \text{Zn}^{2+} + 2\text{e}^-$
 Cathode: $2\text{MnO}_2 + 2\text{NH}_4^+ + 2\text{e}^- \longrightarrow \text{Mn}_2\text{O}_3 + 2\text{NH}_3 + \text{H}_2\text{O}$
 The minimum mass of reactants, if a dry cell is to generate 0.25 A for 9.65 hrs are:
 (Molar mass of Mn = 55 g mol⁻¹, Zn = 65.4 g mol⁻¹)
 (Neglect any other chemical reactions occurring in the cell)
- (A) 2.943 g Zn (B) 3.915 g MnO₂ (C) 7.83 g MnO₂ (D) 0.81 g NH₄⁺



In the above reaction, which of the following is/are correct:

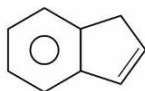
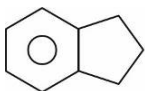
(A) X = n-butane, Y = anti aromatic compound

(B) step-2 is S_N2 reaction

(C) Z is 

(D) The nucleophile in 2nd reaction is $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2^-$

12. Which of the following compounds will not give methane on treatment with CH_3MgI



SPACE FOR ROUGH WORK

SECTION 3

NUMERICAL VALUE TYPE QUESTIONS

This section has 06 Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25)

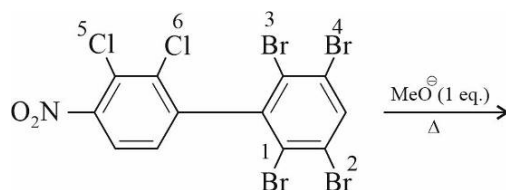
13. Safrole is contained in oil of sassafras and was once used to flavour root of beer. A 2.4 mg sample of safrole was dissolved in 100mg of diphenyl ether. The solution had a freezing point of 25.64°C . The freezing point of pure diphenyl ether is 26.84°C and the freezing point depression constant of diphenyl ether is 8°C/m . The molar mass of safrole in kg/mol is _____.

14. An alloy weighing 2.7 mg of Pb–Ag was dissolved in desired amount of HNO_3 and volume was made 250ml. A silver electrode was dipped in the solution and E_{cell} of the cell $\text{Pt} | \text{H}_2(1\text{bar}) | \text{H}^+(1\text{M}) || \text{Ag}^+ | \text{Ag}$ was 0.5V at 298K. The percentage of lead in the alloy is _____.

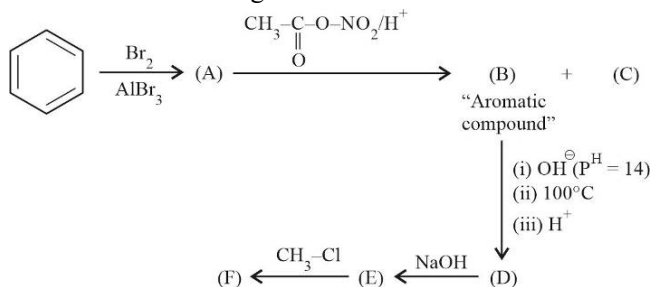
(Given: $E_{\text{Ag}^+|\text{Ag}}^\circ = 0.8\text{V}$, $\frac{2.303RT}{F} = 0.06$, molar mass of Ag = 108 g mol^{-1})

15. X-ray diffraction analysis of crystalline C_{60} (buckminster fullerene) shows that the crystal structure at 300K can be regarded as FCC with a one-molecule basis and unit cell edge length of $4\sqrt{2}\text{ nm}$. The molecules are orientationally disordered due to rotation. The centre to centre distance (in $\text{\AA}$) between nearest neighbour C_{60} molecules in the solid at 300K is _____.

16. For the reaction: $R-X + OH^- \longrightarrow R-OH + X^-$
 Rate law is given as: $\text{rate} = (6 \times 10^{-5})[R-X][OH^-] + [2 \times 10^{-7}][R-X]$. The percentage of $R-X$ react by S_N2 mechanism when $[OH^-] = 0.01M$ is _____.
17. In the following reaction the nucleophile (MeO^-) will displace which of the halogen atom most readily.



18. Consider the following reaction:



The molar mass of product 'F' in $g\ mol^{-1}$ is _____
 (Molar mass of $C = 12, H = 1, O = 16, N = 14\ g\ mol^{-1}$)

SPACE FOR ROUGH WORK

SUBJECT III : MATHEMATICS**66 MARKS****SECTION 1****SINGLE CORRECT ANSWERS TYPE**

This section contains 06 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- If $f(x) = A \sin\left(\frac{\pi x}{2}\right) + B$, $f'\left(\frac{1}{2}\right) = \sqrt{2}$ and $\int_0^1 f(x) dx = \frac{2A}{\pi}$, then the constants A and B are respectively:
 (A) $\frac{\pi}{2}$ and $\frac{\pi}{2}$ (B) $\frac{2}{\pi}$ and $\frac{3}{\pi}$ (C) 0 and $\frac{-4}{\pi}$ (D) $\frac{4}{\pi}$ and 0
- Let $f(x)$ be maximum and $g(x)$ be minimum of $\{x|x|, x^2|x|\}$, then $\int_{-1}^1 (f(x) - g(x)) dx =$
 (A) $\frac{1}{12}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{7}{12}$
- Differential equation of the family of circles touching the line $y = 2$ at $(0, 2)$ is:
 (A) $x^2 + (y-2)^2 + \frac{dy}{dx}(y-2) = 0$ (B) $x^2 + (y-2)\left(2 - 2x\frac{dx}{dy} - y\right) = 0$
 (C) $x^2 + (y-2)^2 + \left(\frac{dx}{dy} + y - 2\right)(y-2) = 0$ (D) None of these
- The equation of the curves through the point $(1, 0)$ and whose slope is $\frac{y-1}{x^2+x}$ is:
 (A) $(y-1)(x+1) + 2x = 0$ (B) $2x(y-1) + x + 1 = 0$
 (C) $x(y-1)(x+1) + 2 = 0$ (D) None of these
- If $\vec{r} = 3\vec{i} + 2\vec{j} - 5\vec{k}$, $\vec{a} = 2\vec{i} - \vec{j} + \vec{k}$; $\vec{b} = \vec{i} + 3\vec{j} - 2\vec{k}$ and $\vec{c} = -2\vec{i} + \vec{j} - 3\vec{k}$ such that $\vec{r} = \lambda\vec{a} + \mu\vec{b} + \nu\vec{c}$, then:
 (A) $\mu, \frac{\lambda}{2}, \nu$ are in A.P. (B) λ, μ, ν are in A.P.
 (C) λ, μ, ν are in H.P. (D) μ, λ, ν are in G.P.
- If $\vec{a}$ and $\vec{b}$ are two unit vectors, then the vector $(\vec{a} + \vec{b}) \times (\vec{a} \times \vec{b})$ is parallel to the vector:
 (A) $\vec{a} - \vec{b}$ (B) $\vec{a} + \vec{b}$ (C) $2\vec{a} - \vec{b}$ (D) $2\vec{a} + \vec{b}$

SPACE FOR ROUGH WORK

SECTION 2**MULTIPLE CORRECT ANSWERS TYPE**

This section consists of 06 Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

7. The value of $\int \frac{dx}{1+e\cos x}$ must be same as:
- (A) $\frac{1}{\sqrt{1-e^2}} \tan^{-1} \left(\sqrt{\frac{1-e}{1+e}} \tan \frac{x}{2} \right) + c$ (e lies between 0 and 1)
- (B) $\frac{2}{\sqrt{1-e^2}} \tan^{-1} \left(\sqrt{\frac{1-e}{1+e}} \tan \frac{x}{2} \right) + c$, (e lies between 0 and 1)
- (C) $\frac{1}{\sqrt{e^2-1}} \log \left| \frac{e + \cos x + \sqrt{e^2-1} \sin x}{1 + e \cos x} \right| + c$, (e is greater than 1)
- (D) $\frac{2}{\sqrt{e^2-1}} \log \left| \frac{e + \cos x + \sqrt{e^2-1} \sin x}{1 + e \cos x} \right| + c$, (e is greater than 1)
8. The area enclosed by the curves $x = a \sin^3 t$ and $y = a \cos^3 t$ is equal to:
- (A) $12a^2 \int_0^{\pi/2} \cos^4 t \sin^2 t dt$
- (B) $12a \int_0^{\pi/2} \cos^2 t \sin^4 t dt$
- (C) $2 \int_{-a}^a (a^{2/3} - x^{2/3})^{3/2} dx$
- (D) $4 \int_0^a (a^{2/3} - x^{2/3})^{3/2} dx$
9. Which one of the following function(s) is/are homogeneous?
- (A) $f(x, y) = \frac{x-y}{x^2+y^2}$
- (B) $f(x, y) = x^{\frac{1}{3}} y^{\frac{2}{3}} \tan^{-1} \frac{x}{y}$
- (C) $f(x, y) = x \left(\ln \sqrt{x^2+y^2} - \ln y \right) + ye^{x/y}$
- (D) $f(x, y) = x \left[\ln \frac{2x^2+y^2}{x} - \ln(x+y) \right] + y^2 \tan \frac{x+2y}{3x-y}$
10. The equation of the curve satisfying the differential equation $y_2(x^2+1) = 2xy_1$ passing through the point (0, 1) and having slope of tangent at $x=0$ as 3 (where y_2 and y_1 represents 2nd and 1st order derivative), then:
- (A) $y = f(x)$ is a strictly increasing function
- (B) $y = f(x)$ is a non-monotonic function
- (C) $y = f(x)$ has three distinct real roots
- (D) $y = f(x)$ has only one negative root

11. The differential equation of the curve for which the initial ordinate (y intercept) of any tangent is equal to the corresponding subnormal:
 (A) is linear (B) is homogeneous of first degree
 (C) has separable variables (D) is second degree
12. Let $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$ and $\vec{c} = \hat{i} + \hat{j} - 2\hat{k}$ be three vectors. A vector in the plane of $\vec{b}$ and $\vec{c}$ whose projection on $\vec{a}$ is of magnitude $\sqrt{\frac{2}{3}}$ is:
 (A) $2\hat{i} + 3\hat{j} - 3\hat{k}$ (B) $2\hat{i} + 3\hat{j} + 3\hat{k}$ (C) $-2\hat{i} - \hat{j} + 5\hat{k}$ (D) $2\hat{i} + \hat{j} + 5\hat{k}$

SPACE FOR ROUGH WORK

SECTION 3

NUMERICAL VALUE TYPE QUESTIONS

This section has 06 Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25)

13. If $f'(x) = 3x^2 \sin \frac{1}{x} - x \cos \frac{1}{x}$, $x \neq 0$, $f(0) = 0$, then the value of $\left[f\left(\frac{1}{\pi}\right) - 2\pi \right]$ is: (where $[.]$ denotes greatest integer functions). Also $f(x)$ is continuous at $x = 0$.
14. The area bounded by the curves $y = x(x-3)^2$ and $y = x$ is _____. (in sq. units)
15. The degree of the differential equation whose general solution is given by $y = (c_1 + c_2)\cos(x + c_3) - c_4 e^{x+c_5}$ where c_1, c_2, c_3, c_4, c_5 are arbitrary constants, is:
16. If $x^3 \frac{dy}{dx} = y^3 + y^2 \sqrt{y^2 - x^2}$ and $y = 1$ when $x = 1$, then $y = 2$ when $x = \dots\dots\dots (x \neq 0)$
17. If $\vec{a}, \vec{b}, \vec{c}$ are three vectors of magnitude $\sqrt{3}, 1, 2$ such that $\vec{a} \times (\vec{a} \times \vec{c}) + 3\vec{b} = 0$. If θ is the angle between $\vec{a}$ and $\vec{c}$, then $\cos^2 \theta =$
18. Let $a, b > 0$ and $\vec{\alpha} = \frac{\hat{i}}{a} + \frac{4\hat{j}}{b} + b\hat{k}$ and $\vec{\beta} = b\hat{i} + a\hat{j} + \frac{1}{b}\hat{k}$ then the maximum value of $\frac{1}{5 + \vec{\alpha} \cdot \vec{\beta}}$ is_____.

SPACE FOR ROUGH WORK

End of JEE Advanced-4 | PAPER-1 | JEE-2022